



PENTECOST UNIVERSITY

WEB TECHNOLOGIES AND DEVELOPMENT

SEMESTER PROJECT TITLE: STANDING DESKS LTD

Date of Submission : 10th December, 2025

Filename : StudentID_ProjectName (e.g. PUIS000111_StandingDesk)

Students must **not** use templates that they have not designed or created in the website assessment. This includes website building applications, free HTML5 website templates, or any software that is available to them to help with the assessment. Students must create their own HTML pages including CSS, JavaScript files and ideally, they will do this through using Notepad++ or a similar text editor.

Introduction

Standing Desks Ltd has been recently set up and is conscious of the health benefits to users. It is looking for an online e-commerce website to help it promote the rising desks and the health benefits to users.

Scenario

The Standing Desk Company (SDC) has approached you and asked you to help them develop a website where they can share information on the different equipment that they have available and sell the health benefits to the users. There will be a Contact Us section on the form for customers to ask for further information. The website should be **visual and viewable on different devices**.

The Standing Desk Company (SDC) has the following basic requirements for the contents of the website which must be **mobile friendly** and include **responsive image** and **text size**:

- **Home Page:**
 - This page will introduce customers to the special offers that are available and should include relevant images of the standing desks and some of the health benefits.
 - The text should be appropriate, and visuals should be used to break up the written content.
 - This page should include a navigation bar (with drop down features), including multimedia elements, header, sections, and footer.
 - Modal popup window that is displayed on top of the current page which provides a “hello” message.

- **Rising Desks:** Examples of the different rising desks that are available e.g., those for laptops on desks or those that are floor standing.
- **Contact Us Page:** This page should have with responsive contact section including first name, last name, and submit button (through email)
- **Health Benefits page:** This page should give the customers information on the health benefits of using the desks.
- **Basket:** Page created which allows the customers to add products to their basket.
- **Mini Apps:** This page should contain all pieces of apps you have created. You must have links to access these apps.

At the initial stage of the development process, you are required to make an HTML/CSS prototype of the website that will clearly show SDC how the final website could work.

Content hasn't been provided. Familiarise yourself with possible types of content by choosing an appropriate organisation of your choice (by using web resources) to help you understand the context in which the company operates. However, do not limit yourself to web-based sources of information. You should also use academic, industry and other sources. Suitable content for your *prototype* can be found on the web e.g. images.

Use Creative Commons (<http://search.creativecommons.org/>) or Wikimedia Commons (http://commons.wikimedia.org/wiki/Main_Page) as a starting point to find content.

Remember the content you include in your site must be licensed for re-use. Do not spend excessive amounts of time researching and gathering content. The purpose is to provide a clear indication of how the final website could look and function. The client would provide the actual content at a later point if they are happy with the website you have proposed.

Task 1– 25 Marks

HTML

The website must be developed using HTML 5 **and** feature a minimum of SIX (6) interlinked pages which can also be viewed on a mobile device. The website must feature the content described above **and** meet the following criteria:

- Research relevant content to inform the design of the website.
- Be usable in TWO (2) different web browsers including being optimised for mobile devices and responsive design.

- Include relevant images of standing desk equipment and the health benefits of using these that you have selected from your research including the use of headers, sections and footers.
- Home Page:
 - Text should be appropriate and visuals should be used to break up the written content.
 - This page should include a navigation bar (with drop down features), including multimedia elements, header, sections, and footer.
 - Responsive design and ONE (1) multimedia element
 - Header ○ Sections
 - Footer (including social media links)
- Rising Desks: Examples of different rising desks that are available e.g. those for laptops on desks or those that are floor standing including THREE (3) floating objects.
- Contact Us Page: This page should have responsive contact section including first name, last name, and submit button (through email).
- Health Benefits Page: with responsive resize design and embedding video and audio for multimedia experiences.
- Basket – Page created which allows the customers to add products to their basket.
- Mini App: Page that links all web tasks or assignments you have created.

Task 2 – 25 Marks

CSS

Create an external CSS file that specifies the design for the website. Each of the HTML pages must link to this CSS file. There should be no use of the style attribute or the <style> element in the website.

The boxes on the home page should include relevant elements such as a slideshow, responsive image grid, transparent overlay text, navigation bar, border radius, box-shadow, hover etc.

Include on-page animated text search to allow customers to search for different products.

Task 3 – 15 Marks

JavaScript

For this task, you are required to integrate **client-side JavaScript** to enhance the interactivity, responsiveness, and functional performance of the Standing Desk Company (SDC) prototype website. The JavaScript implementation must be purposeful, aligned with e-commerce best practices, and demonstrate your competency in front-end scripting.

- Create a JavaScript-driven modal that displays a welcome message on page load with open/close controls.
- Implement dropdown and mobile hamburger menu behaviour using JavaScript event handlers.
- Build a simple add-to-basket system:
 - Add products with name, price, and quantity.
 - Update totals dynamically.
 - Store basket data temporarily using localStorage.
- Add a product search box with animated expansion and dynamic filtering of items.
- Create a JavaScript slideshow/carousel with autoplay and next/previous navigation.
- Validate first name, last name, and email before submission, showing error prompts where needed.
- Add basic JavaScript controls (play/pause) for the video/audio on the Health Benefits page.
- All JavaScript must be placed in an **external .js file** and linked to all pages.

Task 3 – 15 Marks

Test the website and write a test report

You must use the W3C validation service (<http://validator.w3.org/>) to check your HTML **and** CSS code. You should attempt to remove as many non-compliant features as possible. Make sure the *show source* is selected **and** save the output from this tool.

You must also test your website for accessibility using a relevant screen reader **and** note the findings from the testing. You should also describe how the site works for disabled users.

You must test your website on TWO (2) browsers and give a description of the differences between the browsers and the reasons for these.

Investigate the test discrepancies **and** identify and rectify their causes. Produce a summary outlining what you have found supported with screenshots to show the amendments you have made.

Write a short report assessing the suitability of your testing. Provide an evaluation of any outstanding problems and recommendations for how these could be fixed. Explain the role of the W3C.

The written part of your test should not exceed FIVE HUNDRED (500) words.

Task 4 – 35 Marks

Critical evaluation

Write a report of FIVE HUNDRED (500) words evaluating the website you have submitted:

- a) Discuss and evaluate the suitability of the tools, techniques and software used to develop the website.
- b) Evaluate how intuitive interfaces contribute to delivering an improved User Experience.
- c) Appraise TWO (2) accessibility features you have included in your forms.
- d) Recommend improvements for further development of the website in line with the business objectives for growth to increase the product's market share.

Submission requirements

- A word-processed, printed document containing all the written components of the assignment (Tasks 3 and 4), signed, and dated by your tutor/lecturer.
- A copy of all the word-processed material (Tasks 3 and 4) together with the website (Tasks 1 and 2) on appropriate media (CD, USB flash drive, etc.). Ensure that the web pages can be viewed directly from the submitted media. Do not submit any server-side code; it is not required for this assignment.
- The word count for your report is **500 words** for tasks 3 and 4.

Word Count

The word count for this assignment is **1,000 words**. Pay careful attention to the mark allocation for each task so you don't spend too much time on any ONE (1) section.

You **MUST** state the word count in the header of your assignment and declare the number of words on your assignment cover sheet. You **MUST** confirm that you have not exceeded the stipulated word limit by more than 10%.

Any words written beyond the stipulated limit will not be read by the examiners; no marks will be awarded, and no feedback provided for any text beyond the prescribed limit.

In accordance with accepted academic practice, the notion of a word count includes the following without exception:

- All titles or headings that form part of the actual text. This does not include the fly (or cover) page or reference list.
- All words that form the actual essay
- All words forming the titles for figures, tables, and boxes. However, this does not include the text within boxes, tables, or figures themselves.
- All in-text (bracketed) references
- All directly quoted material.